Properties of Matrix Addition and Scalar Multiplication

Let $A,\ B,\ C$ be $m \times n$ matrices and p and q be two non-zero scalars (numbers). Then we have the following properties.

(i) $A + B = B + A$ [Commutative property of matrix addition]

(ii) $A + (B + C) = (A + B) + C$ [Associative property of matrix addition]

(iii) $(pq)A = p(qA)$ [Associative property of scalar multiplication]

(iv) $IA=A$ [Scalar Identity property where I is the unit matrix]

(v) $p(A + B) = pA + pB$ [Distributive property of scalar and two matrices]

(vi) $(p + q)A = pA + qA$ [Distributive property of two scalars with a matrix]

Additive Identity

The null matrix or zero matrix is the identity for matrix addition.

Let A be any matrix.

Then, $A + O = O + A = A$ where O is the null matrix or zero matrix of same order as that of A.

Additive Inverse

If A be any given matrix then $-A$ is the additive inverse of A.

In fact we have $A + (-A) = (-A) + A = O$

Example 3.65 Find the value of $a,\ b,\ c,\ d$ from the following matrix equation.

$$\begin{pmatrix} d & 8 \\ 3b & a \end{pmatrix} + \begin{pmatrix} 3 & a \\ -2 & -4 \end{pmatrix} = \begin{pmatrix} 2 & 2a \\ b & 4c \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -5 & 0 \end{pmatrix}$$

Solution

First, we add the two matrices on both left, right hand sides to get

$$\begin{pmatrix} d+3 & 8+a \\ 3b-2 & a-4 \end{pmatrix} = \begin{pmatrix} 2 & 2a+1 \\ b-5 & 4c \end{pmatrix}$$

Equating the corresponding elements of the two matrices, we have

$$d + 3 = 2 \quad \Rightarrow \quad d = -1$$

$$8 + a = 2a + 1 \quad \Rightarrow \quad a = 7$$

$$3b - 2 = b - 5 \quad \Rightarrow \quad b = \frac{-3}{2}$$

Substituting $a = 7$ in $a - 4 = 4c \ \Rightarrow c = \frac{3}{4}$

Therefore, $a = 7,\ b = -\frac{3}{2},\ c = \frac{3}{4},\ d = -1.$

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